



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Nanyang Business School

WEEK 4

Agentic Automation

Agent

Today's exercise

SHEET 1

customer_emails

Incoming customer emails: address, subject, body

SHEET 2

appointments

The appointment records: a Name column and a DateTime column



The automation

- 1 Reads each email
- 2 Works out what the customer wants
- 3 Updates the appointment record
- 4 Sends a reply

The three requests

1

Check

Tell me my appointment date

2

Change

Move it to a new date

3

Cancel

Remove it

All three arrive as free text. The customer does not fill in a form.

Why rules alone are not enough here

"Can I move my Tuesday slot"



"Tuesday no longer works"



"pls reschedule"



All three are

the same request

Change

- With If and Switch alone, every new wording needs another condition
- The wordings cannot be listed in advance

What agentic automation is

An automation where a language model is given a goal, a set of tools, and the freedom to choose which tool to call and what values to pass it.

You supply

You write the instructions and supply the tools

The model decides

It works out the steps at run time, not at design time

RPA still acts

The model does not click, read sheets, or send email. Activities do all of that

The four parts of an agent

1 Instructions

The system prompt: role, rules, what tools exist

2 Context

The user prompt: this customer's email

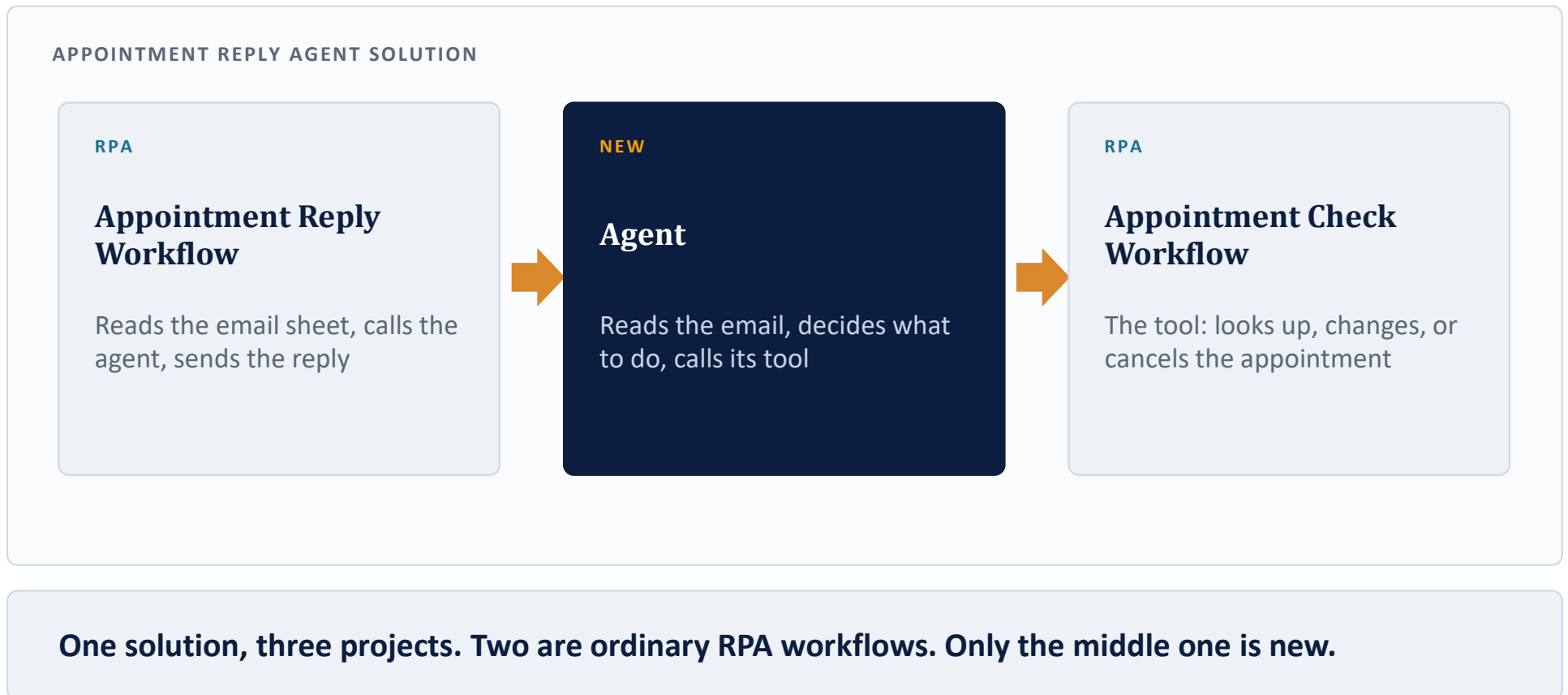
3 Tools

RPA workflows the agent can call

4 Output

What the agent must return. Here, the reply text

How the three projects fit together



System prompt

The standing instruction.

It contains

- The role the agent is playing
- The steps it should follow
- The rules it must not break
- The required output format

1

Fixed

Supplied to you as a text file. You will paste it, not write it.

User prompt

2

Changes on every run

Also supplied to you as a text file.

The part that changes on every run.

Here it carries one customer's email body and email address.

SHAPE OF THE USER PROMPT

Customer Email:

<the email body for this row>

Customer Email Address:

<the address for this row>

Which prompt does a value belong in?

Same on every run

 **System prompt**

Changes for each row of the sheet

 **User prompt**

DIAGNOSTIC WHEN IT MISBEHAVES

Wrong every time

 check the system prompt

Wrong on only some rows

 check the user prompt, or the data

Data Manager

The panel where you declare the named data a project uses. The button looks the same in both project types. The contents differ.

IN AN

Agent project

Inputs

Outputs

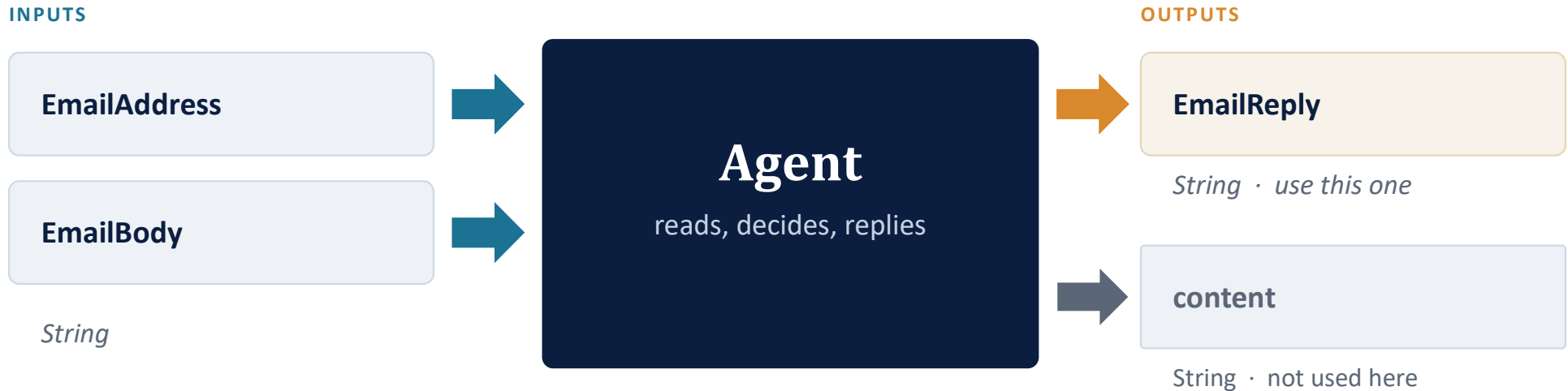
IN AN

RPA workflow

Arguments

Variables

Agent Inputs and Outputs



- Inputs are what the agent needs to be given. Output is what it hands back
- Whatever you declare here becomes the fields the calling workflow has to fill
- The Data Manager shows a content output as well. Use EmailReply for the reply, not content

Arguments in an RPA workflow

Arguments are how data enters and leaves a workflow.



In

Passed in by whoever calls the workflow



Out

Passed back to the caller



In/Out

Both directions. Used sparingly

THE DIFFERENCE

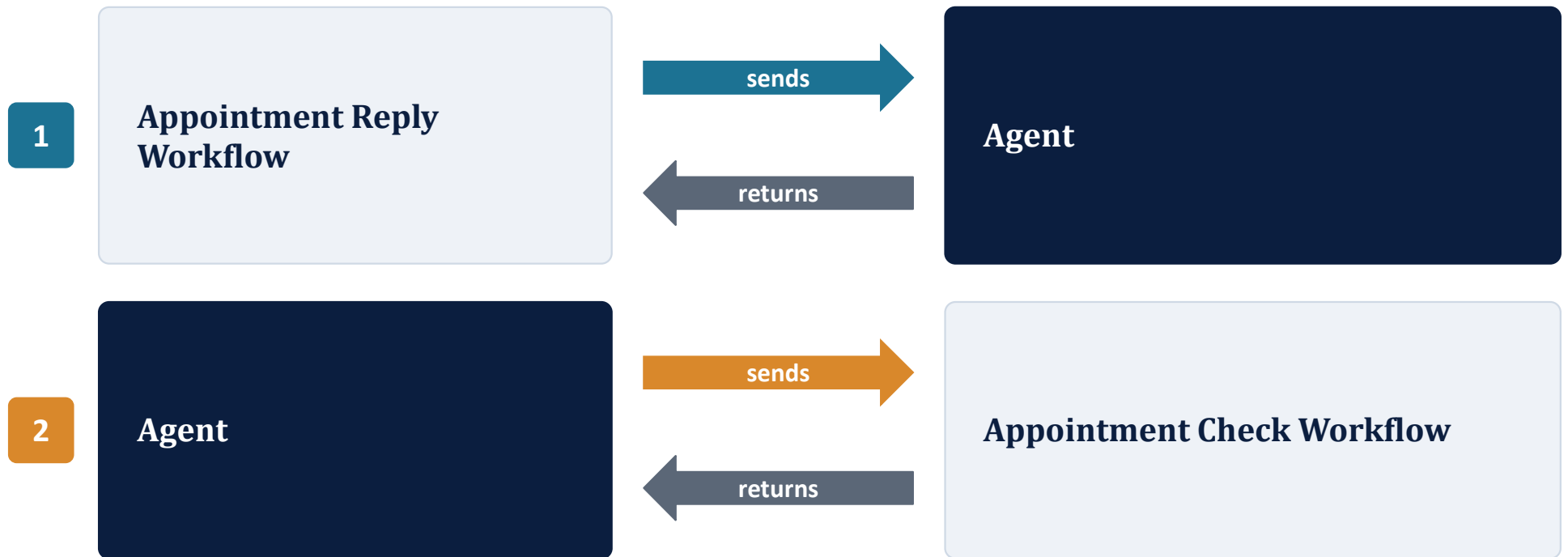
Variables

stay inside one workflow

Arguments

are visible to the caller

The two handoffs



Nothing else talks to anything else. The Appointment Reply Workflow never calls the Appointment Check Workflow directly.

The two handoffs are not the same

1 You wire this one

Appointment Reply Workflow → Agent

- You open Run Job and map each value by hand
- Fixed when you build it
- Every run passes the same two fields

2 Nobody wires this one

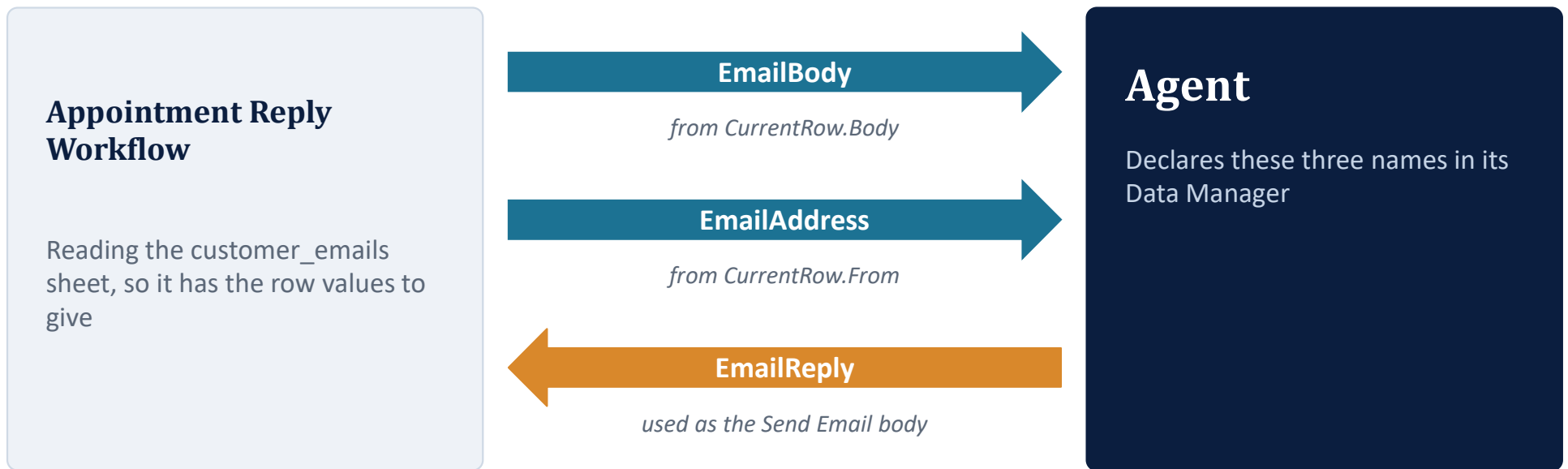
Agent → Appointment Check Workflow

- There is no mapping step anywhere
- The agent reads the tool's inputs and picks the values while it runs
- Different emails produce different values

SO EXACT NAMES ARE NOT REQUIRED

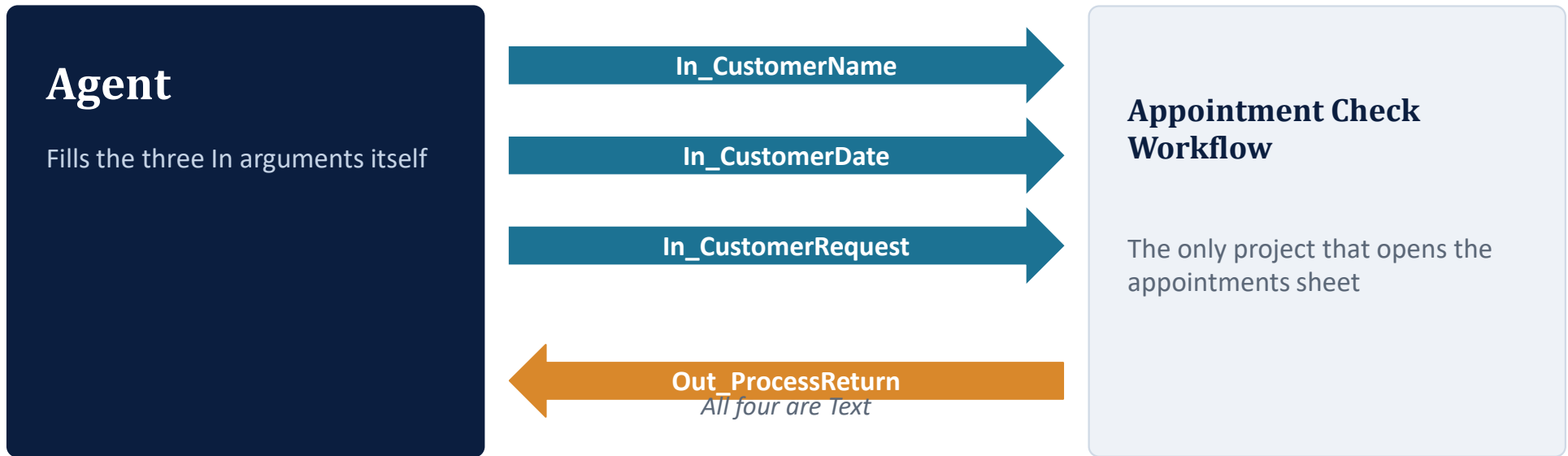
The system prompt asks the agent for CustomerName. The tool declares In_CustomerName. The names differ and it still works, because the agent matches the tool's inputs by meaning.

Handoff 1 — Appointment Reply Workflow to Agent



- All three are String
- CurrentRow.Subject is not passed to the agent. The Appointment Reply Workflow keeps it for the reply subject line

Handoff 2 — Agent to Appointment Check Workflow



OUT_PROCESSRETURN CARRIES BACK ONE OF

the appointment date · Appointment Change · Appointment Cancelled · Appointment not found

Where the agent's three values come from

ONE ROW OF CUSTOMER_EMAILS

From: siti.rahman@example.com
Subject: Appointment

Hi, Tuesday no longer works for me.
Please move my appointment to
14 August 2026, 3pm. Thanks!



In_CustomerName

Siti Rahman



In_CustomerDate

2026-08-14 15:00

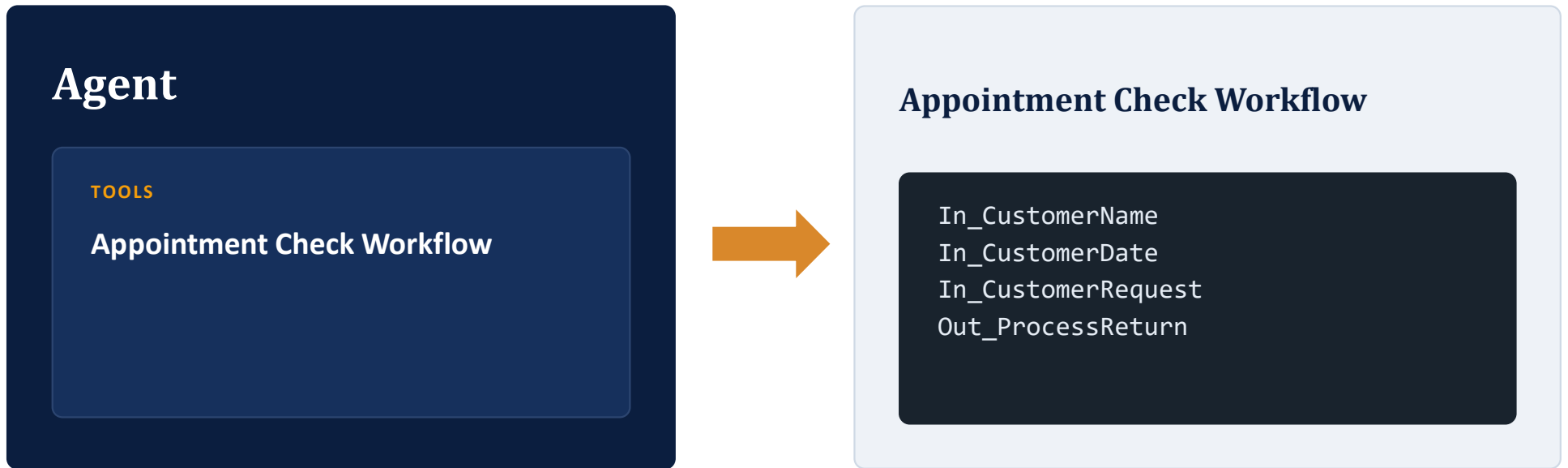


In_CustomerRequest

Change

- The name comes from the email address, not the body. The prompt formats it as FirstName LastName
- The date is rewritten into yyyy-MM-dd HH:mm. The customer never types it that way
- Only the request type is pure judgment. Nothing in the email says the word Change

An RPA workflow as a tool



- Its In arguments become the values the agent fills in
- Its Out argument becomes what the agent reads back

Sending email with Gmail in UiPath

1

Send Email (Google Workspace) calls the Gmail API through a UiPath connection, not through SMTP

2

Connections are per user. Yours is tied to your own Google account and will not work on a classmate's

3

The send sits in the RPA workflow, not in the agent's tools. Email cannot be recalled, so the irreversible action stays under fixed control

THE PATTERN

The agent drafts.

A fixed step sends.

Worth remembering beyond this exercise.

You will connect your own Google account during the build.

What happens next

1

Build

All three projects, then wire them together

2

Run

Replies in your inbox and the appointments sheet updated

3

Ask

Raise a hand the moment something does not match the document

Instructions are in the exercise document. Follow it step by step.